

Maxence Mayrand

Curriculum Vitae | May 2026

email maxence.mayrand@usherbrooke.ca
webpage <https://maxencemayrand.github.io>

Employment

Université de Sherbrooke | Assistant Professor 2022–present
Département de mathématiques

University of Toronto | Postdoctoral Fellow 2019–2022
Department of Mathematics, Geometric Structures Laboratory
Mentor: Prof. Marco Gualtieri

Jesus College (University of Oxford) | Retained Lecturer in Pure Mathematics 2018–2019

Education

University of Oxford | DPhil in Mathematics 2015–2019
Supervisor: Prof. Andrew Dancer

McGill University | BSc Honours Mathematics 2012–2015

Papers

- [1] Mayrand, M., Royer, B., *Complex abelian varieties and quantum error correction: a mathematical framework for GKP codes*
- [2] Maiza, M.M., Mayrand, M., *Lax-Kirchhoff moduli spaces and Hamiltonian 2D TQFT*. [arXiv:2510.23567](#)
- [3] Burelle, J-P., Maiza, M.M., Mayrand, M., *Deformations of quasi-Hamiltonian spaces*. [Differential Geometry and its Applications](#), **103** (2026), 102371
- [4] Crooks, P., Mayrand, M., *Grothendieck-Springer resolutions and TQFTs*. [arXiv:2504.10285](#)
- [5] Crooks, P., Mayrand, M., *The Moore-Tachikawa conjecture via shifted symplectic geometry*. [arXiv:2409.03532](#)
- [6] Crooks, P., Mayrand, M., *Scheme-theoretic coisotropic reduction*. [The Quarterly Journal of Mathematics](#), **77** (2026), no. 1, 221–247
- [7] Mayrand, M., *Shifted coisotropic structures for differentiable stacks*. [Advances in Mathematics](#), **475** (2025)
- [8] Bălibanu, A., Mayrand, M., *Reduction along strong Dirac maps*. [Proceedings of the London Mathematical Society](#), **132** (2026), no. 2, e70123

- [9] Crooks, P., Mayrand, M., *Symplectic reduction along a submanifold*.
[Compositio Mathematica](#), **158** (2022)
- [10] Mayrand, M., *Hyperkähler metrics near Lagrangian submanifolds and symplectic groupoids*.
[J. Reine Angew. Math. \(Crelle's Journal\)](#), **782** (2022)
- [11] Mayrand, M., *Kempf–Ness type theorems and Nahm equations*.
[Journal of Geometry and Physics](#), **136** (2019)
- [12] Mayrand, M., *Stratification of singular hyperkähler quotients*.
[Complex Manifolds](#), **9** (2022)
- [13] Mayrand, M., *Stratified hyperkähler spaces from semisimple Lie algebras*.
[Transformation Groups](#), **25** (2020)
- [14] Mayrand, M., *Particle motion in monopoles and geodesics on cones*.
[Symmetry, Integrability and Geometry: Methods and Applications \(SIGMA\)](#), **10** (2014)

PhD thesis (University of Oxford, 2019)

Stratified hyperkähler spaces and Nahm's equations

PDF available [here](#)

Research teams

- [CRM](#) | Centre de recherches mathématiques
- [CIRGET](#) | Centre interuniversitaire de recherches en géométrie et topologie
- [SAG](#) | Équipe de recherche en structures algébriques et géométriques à Sherbrooke

Research grants

- New Frontiers in Research Fund – Exploration (\$250,000) 2026–2028
Abelian varieties and bosonic codes: using algebraic geometry to protect against quantum errors
With Baptiste Royer
- FRQNT Team Research Project (\$150,000) 2024–2027
Métriques kählériennes spéciales singulières et non-compactes
With Vestislav Apostolov, Julien Keller, and Frédéric Rochon
- NSERC Discovery Grant (\$150,000) 2023–2028
Quantization and hyperkähler geometry
- NSERC Discovery Launch Supplement (\$12,500) 2023–2024

Scholarships and awards

University of Toronto

- F. V. Atkinson Teaching Award June 2021
- NSERC Postdoctoral Fellowship 2019–2021
- FRQNT Postdoctoral Research Scholarship 2019–2021

Pacific Institute for the Mathematical Sciences

- PIMS Postdoctoral Fellowship (declined) 2019–2021

University of Oxford

- Moussouris Scholarship 2015–2019
Full scholarship (tuition fees and living expenses)
Awarded by the University of Oxford
- NSERC CGS D (declined) 2017–2019
Alexander Graham Bell Canada Graduate Scholarships
- NSERC PGS D 2017–2019
Postgraduate Scholarships–Doctoral Program
- FRQNT Doctoral Scholarship (declined) 2017–2019
- FRQNT Master Scholarship 2015–2017

McGill University

- Dean's Honour List May 2015
- First Class Honours in Mathematics May 2015
- Gowrisankaran Prize in Mathematics and Statistics May 2015
- NSERC Undergraduate Student Research Award Summer 2014
- Kamran Undergraduate Award Summer 2013

Teaching

Université de Sherbrooke

- Équations différentielles (MAT528) Winter 2026
- Analyse I (MAT129) Winter 2026
- Théorie quantique (MAT458) Fall 2025
- Théorie quantique (MAT458) Fall 2024
- Analyse II (MAT346) Fall 2024
- Analyse I (MAT129) Winter 2024
- Analyse réelle (MAT189) Winter 2024
- Géométrie différentielle (MAT603) Fall 2023
- Analyse réelle (MAT189) Winter 2023
- Équations différentielles (MAT528) Winter 2023

University of Toronto

- Complex Variables (MAT334) Winter 2022
- Introduction to Combinatorics (MAT344) Fall 2021
- Introduction to Combinatorics (MAT344) Fall 2020
- Concepts in Abstract Mathematics (MAT246) Winter 2020

Jesus College, University of Oxford

- Group Theory Spring 2019
- Groups and Group Actions Winter and Spring 2019
- Linear Algebra II Winter 2019
- Topology Winter 2019

- Linear Algebra I Fall 2018
- Introduction to University Mathematics Fall 2018
- Introduction to Complex Numbers Fall 2018

Mathematical Institute, University of Oxford

Tutor:

- Lie Algebras Spring 2017
- Lie Algebras Fall 2017

Teaching assistant:

- Commutative Algebra Winter 2018
- Representation Theory of Semisimple Lie Algebras Winter 2017
- Lie Algebras Fall 2016

Talks

1. *Lax-Kirchhoff moduli spaces and Hamiltonian 2D TQFT* 5 Dec 2025
The Fields Institute, Geometric Structures Laboratory Seminar
2. *Scheme-theoretic coisotropic reduction* 21 Nov 2024
Gamma seminar
University of Warsaw
3. *The Moore-Tachikawa conjecture via shifted symplectic geometry* 24 Oct 2024
Mathematical physics seminar
Perimeter Institute for Theoretical Physics
4. *The Moore-Tachikawa conjecture via shifted symplectic geometry* 15 Jul 2024
Workshop on Hamiltonian Geometry and Quantization, The Fields Institute
5. *Shifted coisotropic structures and Moore-Tachikawa TQFTs* 13 Mar 2024
CIRGET, Algebraic Geometry Seminar
6. *Shifted coisotropic structures for differentiable stacks* 4 Mar 2024
The Fields Institute, Geometric Structures Laboratory Seminar
7. *Shifted coisotropic structures for differentiable stacks* 28 Feb 2024
University of Göttingen, Higher Structures Seminar
8. *Hyperkähler metrics via deformation theory* 10 Mar 2023
CIRGET, Séminaire de géométrie et topologie
9. *Reduction of Dirac realizations* 4 Jan 2023
2023 Joint Mathematics Meetings, Boston
AMS Special Session on Integrable Systems and Symplectic Group Actions
10. *Reduction along a submanifold* 4 Jan 2023
2023 Joint Mathematics Meetings, Boston
AMS Special Session on Resolutions of Singularities and
Cohomology in Geometry and Representation Theory
11. *Twistor constructions of hyperkähler and hypercomplex structures near complex submanifolds* 3 Dec 2022
2022 CMS Winter Meeting, Toronto
Complex Geometry and Moduli Space

12. *Reduction of Dirac realizations* 17 Nov 2022
Global Poisson Webinar
13. *Dirac reduction and shifted symplectic geometry* 8 Feb 2022
SwissMAP Research Station, Les Diablerets
Differentiable Stacks, Poisson Geometry and related geometric structures
14. *Dirac reduction along a submanifold* 10 Jan 2022
CIRM, Luminy
Lie Theory and Poisson Geometry
15. *Symplectic reduction along a submanifold and the Moore-Tachikawa TQFT* 1 May 2021
AMS Spring Western Sectional Meeting
Geometric and Categorical Methods in Representation Theory
American Mathematical Society
16. *Hyperkähler metrics near Lagrangian submanifolds and symplectic groupoids* 22 Jan 2021
Geometry and Topology Seminar
University of Waterloo
17. *Hyperkähler metrics near Lagrangian submanifolds and symplectic groupoids* 11 Jan 2021
Differential Geometry Research Seminar
University of Hamburg
18. *Hyperkähler realizations of holomorphic Poisson surfaces* 11 Dec 2020
Friday Fish Seminar
Utrecht University
19. *Mini-course on Symplectic Reduction and Geometric Invariant Theory* 3 Dec 2020
2020 Canadian Mathematical Society Winter Meeting
20. *Symplectic reduction along a submanifold and the Moore–Tachikawa TQFT* 3 Sep 2020
Northeastern University
Geometry, Physics, and Representation Theory Seminar
21. *Hyperkähler structures on holomorphic symplectic groupoids* 7 Jun 2020
Workshop on Lie theory and integrable systems in symplectic and Poisson geometry
22. *Hyperkähler structures on symplectic realizations of holomorphic Poisson surfaces* 28 May 2020
School on Geometry and Dynamics of Foliations
CIRM
23. *Kempf-Ness type theorems and Nahm’s equations* 7 Dec 2019
2019 Canadian Mathematical Society Winter Meeting
Algebraic Geometry and Representation Theory
Toronto, Canada
24. *Stratified hyperkähler spaces, GIT, and Nahm’s equations* 5 Dec 2019
Boston College
Geometry and Topology Seminar
25. *Symplectic reduction, GIT quotients, and Nahm’s equations* 21 Nov 2019
McMaster University
Geometry and Topology Seminar

26. *Singular hyperkähler reduction and geometric invariant theory* 11 Nov 2019
University of Toronto
Symplectic Geometry Seminar
27. *Two lectures on Nahm's equations* 30–31 Oct 2019
The Fields Institute
Geometric Structures Laboratory Seminar
28. *Symplectic reduction, GIT quotients, and Nahm's equations* 11 Oct 2019
CIRGET (Centre interuniversitaire de recherches en géométrie et topologie)
Geometry and Topology Seminar
29. *Singular hyperkähler quotients* 16 May 2019
University of Duisburg-Essen, Germany
The Essen Seminar for Algebraic Geometry and Arithmetic
30. *Stratified hyperkähler spaces* 6 Feb 2019
Centre for Quantum Geometry of Moduli Spaces, Aarhus, Denmark
QGM seminars
31. *Hyperkähler quotients and geometric invariant theory* 5 Feb 2019
Centre for Quantum Geometry of Moduli Spaces, Aarhus, Denmark
Introductory seminars
32. *Nahm's equations, groupoids, and Higgs bundles* 25 Jul 2018
(Part II of a mini-course on my thesis work)
The Fields Institute
Geometric Structures Laboratory Seminar
33. *Stratified hyperkähler spaces and Nahm's equations* 24 Jul 2018
(Part I of a mini-course on my thesis work)
The Fields Institute
Geometric Structures Laboratory Seminar
34. *Stratified hyperkähler spaces* 15 Jun 2018
University of Edinburgh
GLEN Algebraic Geometry Seminar
35. *Stratified hyperkähler spaces* 5 Mar 2018
University of Oxford
Geometry and Analysis Seminar
36. *Nahm equations, stratified hyperkähler spaces and the Moore–Tachikawa category.* 30 Mar 2017
ICTS, Tata Institute of Fundamental Research, Bangalore, India
ICTS Special Seminar
37. *Nahm equations and stratified hyperkähler spaces* 28 Mar 2017
Indian Institute of Science, Bangalore, India
School and Workshop on Geometry and Physics of Moduli Spaces